

Week 1: Absolute Value and Inequalities

Jacob Khohayting

Last updated: September 20, 2026

Contents

Inequalities	2
Absolute Value	3
AM-GM Inequality	3
Graphing	4
Practice Problems	6

※ Inequalities

Inequalities deal with relations of variables denoted by four signs: $>$, $<$, $\geq$, $\leq$. For two numbers a and b :

- $a > b$ if a is greater than b , that is, $a - b$ is positive.
- $a < b$ if a is smaller than b , that is, $a - b$ is negative.
- $a \geq b$ if a is greater than or equal to b , that is, $a - b$ is nonnegative.
- $a \leq b$ if a is less than or equal to b , that is, $a - b$ is nonpositive.

Inequalities can sometimes be tricky to solve. So, we will take a look at a few examples!

Solve $4x + 3 > 3x + 4$.

Inequalities work very similar to ordinary systems of equations. So, in this case, subtract $3x$ and 3 on both sides to get $\boxed{x > 1}$.

Solve $\frac{x-8}{x+5} + 4 \geq 3$ where defined.

Once again, they work fairly similar to systems of equations so we can do some manipulation:

$$\frac{x-8}{x+5} + 4 \geq 3 \implies \frac{x-8}{x+5} + 1 \geq 0 \implies \frac{2x-3}{x+5} \geq 0.$$

If this were a system of equation we were solving and instead of the “ $\geq$ ” sign, we had an “ $=$ ” sign, then we know $x + 5 \neq 0$. Otherwise, the equation is undefined. So, we could multiply by $x + 5$ to obtain $2x - 3 = 0$. This doesn’t exactly work in our scenario though. To see why imagine $5 > 2$ which is a true statement. Multiplying by -1 yields $-5 > -2$ which is false! The rule is that we actually have to multiply by -1 if we end up multiplying both sides by a negative number.

In this scenario, we don’t know whether $x + 5$ is greater than 0 or less than. We can obviously ignore $x + 5 = 0$ as then the expression is undefined. So, we have to take cases:

- If $x + 5 > 0$, then multiplying by $x + 5$ yields $2x - 3 \geq 0$. So, $x > -5$ and $x \geq \frac{3}{2}$. Both of these must be satisfied. The stricter one is $x \geq \frac{3}{2}$.
- If $x + 5 < 0$, then multiplying by $x + 5$ forces us to flip the inequality yielding $2x - 3 \leq 0$. So, $x < -5$ and $x \leq \frac{3}{2}$. The stricter one is $x < -5$.

So, $x < -5$ or $x \geq \frac{3}{2}$ are the solution sets. Mathematicians would write this as

$$\boxed{(-\infty, -5) \cup \left[\frac{3}{2}, \infty\right)}.$$

Here, the open bracket’s (and) are used when we have the $<$ or $>$ signs. For instance, $x < -5$, so we used the open bracket. On the other hand, closed bracket’s [and] are used when we have $\geq$ or $\leq$ signs. For instance,

$x \geq \frac{3}{2}$, so we used the closed bracket. In general, when we have infinity, we use the open brackets. We use the $\cap$ notation to denote the word “and” because we want to say both $(-\infty, -5)$ and $[\frac{3}{2}, \infty)$ are solution sets.

※ Absolute Value

The absolute value of a real number x , denoted $|x|$, is the unsigned portion of x . Geometrically, $|x|$ is the distance between x and zero on the real number line.

When x is real, $|x|$ is defined as

$$|x| = \begin{cases} x & \text{for } x \geq 0, \\ -x & \text{for } x \leq 0. \end{cases}$$

Solve $|2x + 3| < 5$ where x is a real number.

Notice, $-5 < 2x + 3 < 5$. Now, we solve this in the same way to the inequalities. So,

$$\begin{aligned} 2x + 3 < 5 &\implies 2x < 2 \implies x < 1, \\ 2x + 3 > -5 &\implies 2x > -8 \implies x > -4. \end{aligned}$$

So, $-4 < x < 1$ which can be written also as $\boxed{(-4, 1)}$.

※ AM-GM Inequality

Convince yourself that $x^2 \geq 0$

For example, $1^2 = 1 > 0$ and $2^2 = 4 > 0$ while $0^2 = 0$. Indeed for negatives, $(-1)^2 = 1 > 0$ and $(-2)^2 = 4 > 0$. This is a fairly basic property but will be crucial in the steps ahead.

Let a and b be non-negative integers. Then,

$$\frac{a+b}{2} \geq \sqrt{ab} = (ab)^{1/2}.$$

Equality holds when $a = b$. That is, the only case in which $\frac{a+b}{2} \geq \sqrt{ab}$ is when $a = b$.

Notice, $(a-b)^2 \geq 0$. So, $a^2 - 2ab + b^2 \geq 0$. Then, $a^2 + 2ab + b^2 \geq 4ab$. So, $(a+b)^2 \geq 4ab$. Taking the square root (everything is positive, so this is allowed), then $a+b \geq 2\sqrt{ab}$. Hence,

$$\frac{a+b}{2} \geq \sqrt{ab},$$

as desired. Notice, equality holds when $(a-b)^2 = 0$ which requires $a-b=0$ and $a=b$!

Indeed, this is extendable, but the proof is too difficult for the current course unfortunately. I am sure you will prove it one day! Here is the more general version:

Let $a_1, a_2, a_3, \dots, a_n$ be non-negative integers. Then,

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq (a_1 a_2 \dots a_n)^{1/n}.$$

Equality holds when $a_1 = a_2 = \dots = a_n$.

Here is one example (or two, I guess)!

Let x be a positive real number. What is the minimum value of

$$x + \frac{9}{x}?$$

By AM-GM Inequality,

$$x + \frac{9}{x} \geq 2\sqrt{x \cdot \frac{9}{x}} = 6.$$

Equality holds when $x = \frac{9}{x}$ requiring $x^2 = 9$ or $x = 3$.

Another way to actually solve this is pretty nice, so I'll share it. If you try graphing it, you will gain some intuition about the minimum. But, you should realize there exists a line $y = k$ which intersects the function

$$f(x) = x + \frac{9}{x}$$

at exactly one point. This value of k is the minimum. When does this occur? Well,

$$x + \frac{9}{x} = k \implies x^2 - kx + 9 = 0.$$

Using the Quadratic Formula, this has one solution, so the discriminant is 0. So,

$$k^2 - 4 \cdot 1 \cdot 9 = 0 \implies k = \pm 6.$$

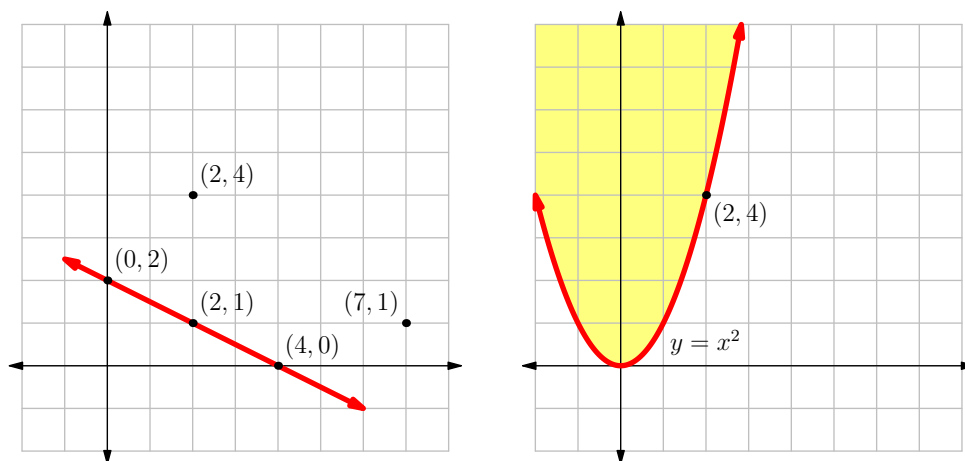
It has to, of course, be 6 since -6 is unattainable when x is a positive number. The next problem is fairly similar. Give it a shot!

Let x be a negative integer. What is the maximum value of

$$x + \frac{9}{x}?$$

✧ Graphing

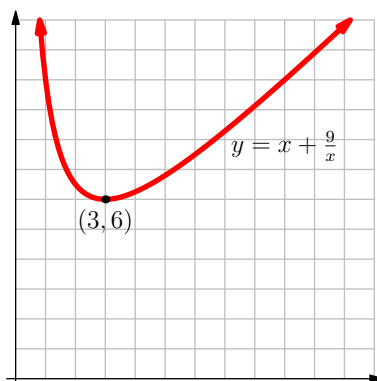
A graph is a visual representation of a function. Below is the coordinate plane:



There are different objects in these diagrams, so let's inspect them:

- In the coordinate plane, there are two axes. The y -axis goes vertically and is the black line above while the x -axis goes horizontally and is the black line below. The axes are like number lines going from $-\infty$ to ∞ .
- Functions are generally defined as $y = f(x)$ where $f(x)$ is some expression in-terms of x . A graph is plotted with each point being (x, y) .
- Points are labeled with dots. Things like $(2, 4)$ are points, for example.
- Lines can be drawn as straight lines. The line shown above is $\frac{x}{4} + \frac{y}{2} = 1$. Every point on the line must satisfy that equation. For example, $(0, 2)$, $(2, 1)$, and $(4, 0)$ all satisfy the equation.
- In the diagram on the right, there is a quadratic function $y = x^2$ graphed in the red. The yellow region denotes the region where $y > x^2$.
- The roots of an equation or the solutions are when the graph intersects the x -axis as y would inevitably be 0. For example, $y = x^2 = 0$ when $x = 0$ which is when it intersects the x -axis. On the otherhand, $\frac{x}{4} + \frac{y}{2} = 1$. Here, when $y = 0$ then $\frac{x}{4} = 1$ so $x = 4$ which we can see is the root.

Graphing can give us insight into various equations and how things work. Things that are visual are generally very helpful. We will utilize graphing a lot in the problem set. Here is one example from before!



Some problems are difficult because they may be hard to graph. So, we will work on graphing things and its applications!

✧ Practice Problems

1. Fifteen integers $a_1, a_2, a_3, \dots, a_{15}$ are arranged in order on a number line. The integers are equally spaced and have the property that

$$1 \leq a_1 \leq 10, 13 \leq a_2 \leq 20, \text{ and } 241 \leq a_{15} \leq 250.$$

What is the sum of digits of a_{14} ?

2. For how many integer values of x is $|2x| \leq 7\pi$?
3. Connor is thinking of a two-digit number n , which satisfies the following properties:
- If $n > 70$, then n is a perfect square.
 - If $n > 40$, then n is prime.
 - If $n < 80$, then the sum of the digits of n is 14.

What is Connor's number?

4. Suppose a , b , and c are nonzero real numbers, and $a + b + c = 0$. What are the possible value(s) for

$$\frac{a}{|a|} + \frac{b}{|b|} + \frac{c}{|c|} + \frac{abc}{|abc|}?$$

5. How many ordered pairs of real numbers (x, y) satisfy the following system of equations?

$$\begin{aligned} x + 3y &= 3 \\ ||x| - |y|| &= 1 \end{aligned}$$

6. Let $a > 5$. Find the smallest possible value of $a + \frac{4}{a-5}$.
7. A dartboard is the region $\mathcal{B}$ in the coordinate plane consisting of points (x, y) such that $|x| + |y| \leq 8$. A target $\mathcal{T}$ is the region where $(x^2 + y^2 - 25)^2 \leq 49$. A dart is thrown at a random point in $\mathcal{B}$. What is the probability that the dart lands in $\mathcal{T}$?
8. What is the area of the region in the coordinate plane defined by the inequality

$$||x| - 1| + ||y| - 1| \leq 1?$$

9. What is the maximum value of $\frac{(2^t - 3t)t}{4^t}$ for real values of t ?
10. The length of the interval of solutions of the inequality $a \leq 2x + 3 \leq b$ is 10. What is ba ?
11. How many ordered pairs (x, y) of real numbers satisfy the following system of equations?

$$\begin{aligned} x^2 + 3y &= 9 \\ (|x| + |y| - 4)^2 &= 1 \end{aligned}$$

12. What is the least possible value of

$$(x+1)(x+2)(x+3)(x+4) + 2026$$

where x is a real number?

13. What is the area of the region enclosed by the graph of the equation $x^2 + y^2 = |x| + |y|$?